

EXHIBIT 18

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON DC**

**Before the Honorable Clark S. Cheney
Chief Administrative Law Judge**

In the Matter of

**CERTAIN SEMICONDUCTOR DEVICES,
PRODUCTS CONTAINING SAME, AND
COMPONENTS THEREOF**

Investigation No. 337-TA-1500

**COMPLAINANT GLOBALFOUNDRIES U.S. INC.'S SECOND SUPPLEMENTAL
OBJECTIONS AND RESPONSES TO RESPONDENTS' FIRST SET OF
INTERROGATORIES (NOS. 1-73)**

Pursuant to the United States International Trade Commission's Rules of Practice and Procedure, 19 C.F.R. §§ 210.27 and 210.29, Complainant GlobalFoundries U.S. Inc. ("GlobalFoundries" or "Complainant") hereby further respond to Respondents Tower Semiconductor Ltd.; Tower Partners Semiconductor, Ltd.; Tower Semiconductor Italy S.r.l.; Tower US Holdings, Inc.; Tower Semiconductor San Antonio, Inc.; Tower Semiconductor NPB Holdings, Inc.; Tower Semiconductor Newport Beach, Inc., and Newport Fab LLC's (collectively "Tower" or "Respondents") First Set of Interrogatories Nos. 1-73.

GlobalFoundries's responses are based on information currently available to GlobalFoundries. GlobalFoundries reserves all rights to supplement, revise, and/or amend these responses, should additional information become available through the discovery process or by other means. In responding to these Interrogatories, GlobalFoundries does not waive any objection on the grounds of privilege, confidentiality, competency, relevance, materiality, authenticity, admissibility of the information contained in these responses, or any other objection.

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GlobalFoundries responds as follows: An invention disclosure form relating to the '425 patent was submitted on January 1, 2013. An invention disclosure form relating to the '167 patent and the '546 patent was submitted on June 6, 2014. An invention disclosure form relating to the '748 patent was submitted on June 21, 2016.

INTERROGATORY NO. 27:

Provide all factual and legal bases for GlobalFoundries' contention that the Asserted Claims are eligible for patenting under 35 U.S.C. § 101.

RESPONSE TO INTERROGATORY NO. 27:

GlobalFoundries incorporates all of its General Objections and reservations of rights as if set forth herein. GlobalFoundries objects to this Interrogatory as a premature contention interrogatory because Ground Rule 4.4.2 provides for contention interrogatory responses in accordance with the deadlines that will be set forth in the Procedural Schedule. GlobalFoundries further objects to this Interrogatory as overbroad and unduly burdensome because it seeks "all factual and legal bases" for patent-eligibility contentions across the Asserted Claims without reasonable limitation and is therefore disproportionate to the needs of this Investigation. GlobalFoundries further objects to this Interrogatory as improper burden-shifting because Tower bears the burden for any patent-eligibility defense under 35 U.S.C. § 101 and Tower has not yet provided any Invalidity Contentions. GlobalFoundries further objects to this Interrogatory as calling for a legal conclusion rather than factual information. GlobalFoundries further objects to this Interrogatory to the extent it seeks information subject to attorney-client privilege, the work product doctrine, and any other applicable protections.

Subject to and without waiving the foregoing objections and General Objections above: Pursuant to Ground Rule 4.4.2, GlobalFoundries will respond to contention interrogatories in accordance with the deadlines set forth in the Procedural Schedule.

INTERROGATORY NO. 28:

Provide all factual and legal bases for GlobalFoundries' contention that the Asserted Claims are valid under pre-AIA 35 U.S.C. § 102 and/or AIA 35 U.S.C. § 102, where applicable.

RESPONSE TO INTERROGATORY NO. 28:

GlobalFoundries incorporates all of its General Objections and reservations of rights as if set forth herein. GlobalFoundries objects to this Interrogatory as a premature contention interrogatory because Ground Rule 4.4.2 provides for contention interrogatory responses in accordance with the deadlines that will be set forth in the Procedural Schedule. GlobalFoundries further objects to this Interrogatory as improper burden-shifting because the invalidity burden rests with Respondents who have not yet provided any Invalidity Contentions. GlobalFoundries further objects to this Interrogatory to the extent it seeks information subject to attorney-client privilege, the work product doctrine, and any other applicable protections. GlobalFoundries further objects to this Interrogatory as overbroad and unduly burdensome because it seeks "all factual and legal bases" without reasonable limitation and is therefore disproportionate to the needs of this Investigation.

Subject to and without waiving the foregoing objections and the General Objections above, GlobalFoundries responds as follows: Pursuant to Ground Rule 4.4.2, GlobalFoundries will respond to contention interrogatories in accordance with the deadlines set forth in the Procedural Schedule.

FIRST SUPPLEMENTAL RESPONSE TO INTERROGATORY NO. 28 (July 30, 2026):

Subject to and without waiving the foregoing objections and the General Objections above, GlobalFoundries responds as follows:

GlobalFoundries responds to Tower's preliminary invalidity contentions. Tower bears the burden of proving invalidity by clear and convincing evidence and must establish that every

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limitation of every asserted claim is disclosed by or rendered obvious in view of the prior art. GlobalFoundries does not concede Tower's claim constructions, priority-date assertions, prior-art-status assertions, or characterizations of the cited references. This response is based on Tower's current contentions and the present record. GlobalFoundries preserves all validity arguments and reserves the right to respond to any amended, supplemented, or otherwise modified contentions.

I. U.S. Patent No. 8,330,235 ("235 Patent")

Tower has failed to meet its burden to establish by clear and convincing evidence that any prior art reference anticipates any asserted claim of the '235 patent. To anticipate, a single prior art reference must disclose each and every limitation of a claim, arranged as in the claim. Tower's five claim charts (Exhibits A1 through A5) each fail to establish that the charted reference discloses all limitations of the claims to which it is mapped.

Tower has failed to meet its burden to establish by clear and convincing evidence that Shimizu '954 discloses: a silicon substrate; a transistor comprising source/drain regions and a gate electrode on the substrate; a nickel silicide layer on the gate electrode and source/drain regions, wherein the nickel silicide layer comprises platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode, from between about 8 wt.% and about 12 wt.% at the silicon interface; a nickel silicide layer having a thickness of about 100 Å to about 150 Å; a silicon substrate comprising a strained silicon germanium (Si-Ge) layer on a silicon layer; a stress liner on the nickel silicide layer; an oxide or oxynitride layer on the stress liner; a stress liner comprising stressed silicon nitride; a P-channel transistor and an N-channel transistor, each comprising source/drain regions

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and a gate electrode on the silicon substrate, a first nickel silicide layer on the gate electrode and source/drain regions of the P-channel transistor, a second nickel silicide layer on the gate electrode and source/drain regions of the N-channel transistor, each of the first and second nickel silicide layers comprising platinum having a concentration gradient from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the transistors, from between about 8 wt.% and about 12 wt.% at the silicon interface; a first stress liner on the P-channel transistor and a second stress liner on the N-channel transistor; a first stress liner exhibiting high compressive stress and a second stress liner exhibiting high tensile stress; nickel silicide layers each having a thickness of about 100 Å to about 220 Å; nickel silicide layers each having a thickness of about 100 Å to about 150 Å; a silicon substrate comprising a strained Si-Ge layer on a silicon layer; an oxide or oxynitride layer on the first and second stress liners; and first and second stress liners comprising stressed silicon nitride.

For example, Tower has not identified a single passage establishing that Shimizu '954 discloses a nickel silicide layer comprising platinum having the asserted composition gradient—i.e., platinum increasing in amount in a direction away from the source/drain regions and gate electrode, from between about 8 wt.% and about 12 wt.% at the silicon interface. Tower likewise has not identified any Shimizu '954 citation for the asserted strained Si-Ge substrate limitation.

For the stress-liner limitations, Tower's cited Shimizu '954 passages describe dual silicon nitride films (e.g., silicon nitride films 13 and 14) positioned "on top of a silicide layer 12" for tensile and compressive stress tuning of NMOS and PMOS devices. *See, e.g.,* Shimizu '954 at 8:37-9:16, 10:47-12:14, Fig. 1. Tower has not shown, however, that the "silicide layer 12" in

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Shimizu '954 is the asserted nickel silicide layer comprising platinum having the claimed composition gradient. Tower's cited passages do not identify the composition of silicide layer 12, and Tower has not established that Shimizu '954's cited stress-liner arrangement is combined, within Shimizu '954 itself, with the asserted platinum-gradient nickel silicide structure in the arrangement the claims require.

Tower has likewise failed to meet its burden to establish by clear and convincing evidence that Akiyama '559 discloses: a silicon substrate; a transistor comprising source/drain regions and a gate electrode on the substrate; a nickel silicide layer on the gate electrode and source/drain regions, wherein the nickel silicide layer comprises platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode, from between about 8 wt.% and about 12 wt.% at the silicon interface; a nickel silicide layer having a thickness of about 100 Å to about 150 Å; a silicon substrate comprising a strained silicon germanium (Si-Ge) layer on a silicon layer; a stress liner on the nickel silicide layer; an oxide or oxynitride layer on the stress liner; a stress liner comprising stressed silicon nitride; a P-channel transistor and an N-channel transistor, each comprising source/drain regions and a gate electrode on the silicon substrate, a first nickel silicide layer on the gate electrode and source/drain regions of the P-channel transistor, a second nickel silicide layer on the gate electrode and source/drain regions of the N-channel transistor, each of the first and second nickel silicide layers comprising platinum having a concentration gradient from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the transistors, from between about 8 wt.% and about 12 wt.% at the silicon interface; a first stress liner on the P-channel transistor and

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a second stress liner on the N-channel transistor; a first stress liner exhibiting high compressive stress and a second stress liner exhibiting high tensile stress; nickel silicide layers each having a thickness of about 100 Å to about 220 Å; nickel silicide layers each having a thickness of about 100 Å to about 150 Å; a silicon substrate comprising a strained Si-Ge layer on a silicon layer; an oxide or oxynitride layer on the first and second stress liners; and first and second stress liners comprising stressed silicon nitride.

Tower has not identified a cited Akiyama '559 passage establishing the asserted platinum composition-gradient relationship—i.e., a nickel silicide layer comprising platinum having a composition gradient with platinum increasing in amount in a direction away from the source/drain regions and gate electrode, from between about 8 wt.% and about 12 wt.% at the silicon interface. Tower's obviousness theory does not supply an anticipatory disclosure in the primary reference, and Tower has not identified a single-reference disclosure of the asserted platinum-gradient relationship in Akiyama '559.

Tower has not shown by clear and convincing evidence that Akiyama '559 discloses the asserted stress-liner arrangement (i.e., a stress liner on the nickel silicide layer, an oxide or oxynitride layer on the stress liner, a first stress liner exhibiting high compressive stress on the P-channel transistor and a second stress liner exhibiting high tensile stress on the N-channel transistor, or stress liners comprising stressed silicon nitride).

Tower cites Akiyama '559, paragraph [0037] and Figure 5A, for a "Ni(Pt) alloy layer" deposited with a thickness of "about 10nm." *See* Tower Preliminary Invalidity Contentions, § V.A.4.b "Silicide Thickness" grouping. This citation describes an as-deposited metal-alloy layer thickness—not a post-anneal nickel silicide layer thickness. Silicidation fundamentally changes layer thickness and composition: the metal-alloy layer reacts with the underlying silicon during

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annealing to form a silicide, and the resulting silicide thickness depends on the stoichiometry of the reaction, the anneal temperature, and the extent of silicon consumption. Tower has not shown that a 10 nm deposited Ni(Pt) alloy film produces, after silicidation and anneal, a nickel silicide layer having the specific claimed thickness of about 100 Å to about 150 Å (or about 100 Å to about 220 Å).

Likewise, Tower has not shown that Akiyama '559's Si-Ge disclosure is combined, within Akiyama '559 itself, with a platinum-gradient nickel silicide structure in the arrangement the claims require. Tower's citation to passages in Akiyama '559 discussing Si-Ge does not, without more, establish that Akiyama '559 discloses the specific combination of limitations as arranged in the claims.

Tower has failed to meet its burden to establish by clear and convincing evidence that Choo '406 discloses: a silicon substrate; a transistor comprising source/drain regions and a gate electrode on the substrate; a nickel silicide layer on the gate electrode and source/drain regions, wherein the nickel silicide layer comprises platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode, from between about 8 wt.% and about 12 wt.% at the silicon interface; a nickel silicide layer having a thickness of about 100 Å to about 150 Å; a silicon substrate comprising a strained silicon germanium (Si-Ge) layer on a silicon layer; a stress liner on the nickel silicide layer; an oxide or oxynitride layer on the stress liner; a stress liner comprising stressed silicon nitride; a P-channel transistor and an N-channel transistor, each comprising source/drain regions and a gate electrode on the silicon substrate, a first nickel silicide layer on the gate electrode and source/drain regions of the P-channel transistor, a second nickel silicide layer on the gate

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electrode and source/drain regions of the N-channel transistor, each of the first and second nickel silicide layers comprising platinum having a concentration gradient from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the transistors, from between about 8 wt.% and about 12 wt.% at the silicon interface; a first stress liner on the P-channel transistor and a second stress liner on the N-channel transistor; a first stress liner exhibiting high compressive stress and a second stress liner exhibiting high tensile stress; nickel silicide layers each having a thickness of about 100 Å to about 220 Å; nickel silicide layers each having a thickness of about 100 Å to about 150 Å; a silicon substrate comprising a strained Si-Ge layer on a silicon layer; an oxide or oxynitride layer on the first and second stress liners; and first and second stress liners comprising stressed silicon nitride.

Tower has not identified a cited Choo '406 passage establishing the asserted platinum composition-gradient relationship—i.e., a nickel silicide layer comprising platinum having a composition gradient with platinum increasing in amount in a direction away from the source/drain regions and gate electrode, from between about 8 wt.% and about 12 wt.% at the silicon interface. Tower's obviousness theory does not supply an anticipatory disclosure in the primary reference, and Tower has not identified a single-reference disclosure of the asserted platinum-gradient relationship in Choo '406.

For the stress-liner limitations, Tower's cited Choo '406 passages describe a stress liner made of "SiCN, SiN, SiON, SiBN, SiO₂, SiC, SiCH, SiCOH or a composite layer of these materials" (Choo '406 at [0024]) with the metal silicide layer 117 positioned under the stress liner 130. Choo '406 at [0028]. Tower has not shown, however, that the metal silicide layer 117 in Choo '406 is a nickel silicide layer having the asserted platinum composition gradient. Tower

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has therefore not established the asserted relationship between the stress liner and the specific claimed nickel-platinum-silicide structure.

Tower has failed to meet its burden to establish by clear and convincing evidence that Lee '487 discloses: a silicon substrate; a transistor comprising source/drain regions and a gate electrode on the substrate; a nickel silicide layer on the gate electrode and source/drain regions, wherein the nickel silicide layer comprises platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode, from between about 8 wt.% and about 12 wt.% at the silicon interface; a nickel silicide layer having a thickness of about 100 Å to about 150 Å; a silicon substrate comprising a strained silicon germanium (Si-Ge) layer on a silicon layer; a stress liner on the nickel silicide layer; an oxide or oxynitride layer on the stress liner; a stress liner comprising stressed silicon nitride; a P-channel transistor and an N-channel transistor, each comprising source/drain regions and a gate electrode on the silicon substrate, a first nickel silicide layer on the gate electrode and source/drain regions of the P-channel transistor, a second nickel silicide layer on the gate electrode and source/drain regions of the N-channel transistor, each of the first and second nickel silicide layers comprising platinum having a concentration gradient from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the transistors, from between about 8 wt.% and about 12 wt.% at the silicon interface; a first stress liner on the P-channel transistor and a second stress liner on the N-channel transistor; a first stress liner exhibiting high compressive stress and a second stress liner exhibiting high tensile stress; nickel silicide layers each having a thickness of about 100 Å to about 220 Å; nickel silicide layers each having a thickness of about 100 Å to about 150 Å; a

silicon substrate comprising a strained Si-Ge layer on a silicon layer; an oxide or oxynitride layer on the first and second stress liners; and first and second stress liners comprising stressed silicon nitride.

Tower has not identified a cited Lee '487 passage establishing the asserted platinum composition-gradient relationship—i.e., a nickel silicide layer comprising platinum having a composition gradient with platinum increasing in amount in a direction away from the source/drain regions and gate electrode, from between about 8 wt.% and about 12 wt.% at the silicon interface. Tower's obviousness theory does not supply an anticipatory disclosure in the primary reference, and Tower has not identified a single-reference disclosure of the asserted platinum-gradient relationship in Lee '487.

Tower's own contentions do not identify Lee '487 as independently disclosing the strained Si-Ge substrate limitation or the stress-liner limitations. Tower has failed to identify a cited Lee '487 passage establishing either of these limitation sets, and Tower's alternative obviousness combinations do not supply a single-reference anticipatory disclosure.

Tower cites Lee '487, paragraph [0022], for a Ni(Pt) alloy layer that is “preferably between about 50 Å and 300 Å thick” over the source and drain regions and the gate. Tower Preliminary Invalidity Contentions, § V.A.3.b “Silicide Thickness” grouping. This passage describes an as-deposited metal-alloy thickness range spanning well below and well above the claimed post-silicidation NiSi thickness ranges (100–150 Å and 100–220 Å). Tower has not shown that this broad, generic deposited-metal range establishes the specific claimed post-anneal silicide-thickness limitations. Moreover, Tower has not established that Lee '487 discloses the platinum-composition-gradient relationship in the first instance, which is a prerequisite for any thickness limitation to attach to the claimed structure. Tower's cited passage at most describes a

deposited-metal thickness for a generic Ni(Pt) alloy film, not the thickness of a post-silicidation nickel silicide layer having the asserted platinum gradient.

Tower has failed to meet its burden to establish by clear and convincing evidence that Ku '182 discloses: a silicon substrate; a transistor comprising source/drain regions and a gate electrode on the substrate; a nickel silicide layer on the gate electrode and source/drain regions, wherein the nickel silicide layer comprises platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode, from between about 8 wt.% and about 12 wt.% at the silicon interface; a nickel silicide layer having a thickness of about 100 Å to about 150 Å; a silicon substrate comprising a strained silicon germanium (Si-Ge) layer on a silicon layer; a stress liner on the nickel silicide layer; an oxide or oxynitride layer on the stress liner; a stress liner comprising stressed silicon nitride; a P-channel transistor and an N-channel transistor, each comprising source/drain regions and a gate electrode on the silicon substrate, a first nickel silicide layer on the gate electrode and source/drain regions of the P-channel transistor, a second nickel silicide layer on the gate electrode and source/drain regions of the N-channel transistor, each of the first and second nickel silicide layers comprising platinum having a concentration gradient from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the transistors, from between about 8 wt.% and about 12 wt.% at the silicon interface; a first stress liner on the P-channel transistor and a second stress liner on the N-channel transistor; a first stress liner exhibiting high compressive stress and a second stress liner exhibiting high tensile stress; nickel silicide layers each having a thickness of about 100 Å to about 220 Å; nickel silicide layers each having a thickness of about 100 Å to about 150 Å; a

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silicon substrate comprising a strained Si-Ge layer on a silicon layer; an oxide or oxynitride layer on the first and second stress liners; and first and second stress liners comprising stressed silicon nitride.

Tower has not identified a single cited Ku '182 passage establishing the asserted platinum composition-gradient relationship, the asserted silicide-thickness range, the asserted strained Si-Ge substrate, or the asserted stress-liner arrangement. At most, Tower articulates a separate obviousness theory in which Ku '182 is supplemented by other references for each of the core limitations of the asserted claims. Tower has not shown that Ku '182 itself discloses the asserted structure, and Tower has failed to meet its burden to establish that Ku '182 anticipates any asserted claim of the '235 patent.

Tower has failed to meet its burden to establish by clear and convincing evidence that the asserted claims are obvious. Tower purports to identify various disparate references allegedly disclosing certain claimed features. But Tower has not shown that any specific combination of references renders any specific asserted claim of the '235 patent obvious.

Tower's obviousness contentions are structured as element-by-element "groupings" of references—collections of disparate prior art references that Tower asserts, individually, disclose a particular claim limitation. Tower Preliminary Invalidity Contentions § V.A.3. A person of ordinary skill in the art ("POSITA") considering the prior art must have had a reason to combine specific teachings in the claimed manner, with a reasonable expectation of success and operability. Tower never explains how any specific combination of references would be made; Tower never identifies the particular combination it will rely upon for any given claim; Tower never explains how any combination would result in an operable semiconductor device meeting all limitations simultaneously; and Tower never provides the required particularized motivation

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to combine any specific set of references. Tower's approach reduces to the impermissible assertion that each feature was independently "well known" or "standard"—a hindsight-driven framework that cannot substitute for the particularized motivation-to-combine showing required by law.

Tower's "NiPt and Gradients" grouping addresses the nickel-silicide and platinum-composition-gradient limitations of claims 1 and 10—requiring a nickel silicide layer comprising platinum having a composition gradient with platinum increasing in amount in a direction away from the source/drain regions and gate electrode, from between about 8 wt.% and about 12 wt.% at the silicon interface. Tower's boilerplate conclusion states: "Nickel platinum silicides are standard and a POSITA would have been motivated to incorporate into any device, as demonstrated by the variety of the prior art references that disclose use of nickel platinum silicides and a gradient of Pt, including those discussed above." Tower Preliminary Invalidity Contentions § V.A.3.a. Tower has failed to carry its burden as to this grouping. Tower has not explained how the teachings of any reference would be applied by a POSITA to produce the specific claimed structure: a single nickel silicide layer having a composition gradient from between about 8 wt.% and about 12 wt.% platinum at the silicon interface, with platinum increasing toward the opposite surface. For example, Corni 1993's description of platinum redistribution in Ni₂Si does not address the formation of a graded platinum distribution through a single NiSi-phase layer, the thermal budget and anneal-sequencing conditions necessary to produce such a graded (rather than uniform or reversed) distribution, or the phase stability of NiSi at the elevated anneal temperatures required to drive platinum redistribution. Tower has not identified how a POSITA would control the anneal parameters to achieve the specific 8–12 wt.% starting concentration at the silicon interface while maintaining a monotonically increasing

gradient toward the outer surface, or how a POSITA would expect the platinum distribution to remain graded (rather than homogenizing) during subsequent processing. Tower's bare assertion that NiPt silicides are "standard" does not address these specific process-integration challenges and does not establish a reasonable expectation of success in producing the claimed gradient.

Tower's "Silicide Thickness" grouping addresses the nickel silicide layer thickness limitations: about 100 Å to about 150 Å (claims 5 and 16) and about 100 Å to about 220 Å (claim 15). For example, Tower cites Akiyama '559, paragraph [0037] and Figure 5A, for a Ni(Pt) alloy layer deposited with a thickness of "about 10nm," and Lee '487, paragraph [0022], for a Ni(Pt) alloy layer "preferably between about 50 Å and 300 Å." Tower also cites Okuno '767, a non-charted secondary reference. Tower states: "Different silicide thicknesses were well known and well established in the prior art. . . . A POSITA would have been motivated to incorporate into silicides of thinner or thicker thicknesses into any device." Tower Preliminary Invalidity Contentions § V.A.3.b.

Tower has failed to carry its burden as to this grouping. An as-deposited metal-alloy thickness is not the same physical parameter as a post-anneal silicide-layer thickness. During silicidation, the deposited metal reacts with underlying silicon; the resulting silicide layer has a different thickness than the as-deposited metal due to silicon consumption and volumetric changes associated with silicide formation. Tower has not shown that citing a deposited Ni(Pt) metal thickness of "about 10nm" (Akiyama '559) or a range of "about 50 Å and 300 Å" (Lee '487) establishes the specific claimed post-silicidation NiSi thickness ranges of about 100-150 Å or about 100-220 Å. Tower has not explained the relationship between deposited-metal thickness and post-anneal silicide thickness for Ni(Pt) alloys, has not identified the stoichiometric conversion factor, and has not shown how a POSITA would arrive at the specific claimed post-

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silicidation thickness range from these deposited-metal starting points with a reasonable expectation of success. Tower's "well known" boilerplate does not address this specific technical gap.

Tower's "SiGe" grouping addresses claims 6 and 17, which require a silicon substrate comprising a strained silicon germanium (Si-Ge) layer on a silicon layer. Tower has failed to carry its burden as to this grouping. Tower has not explained how a POSITA would integrate a strained Si-Ge substrate with the specific platinum-gradient nickel silicide structure and etch-resistance objective addressed by the '235 patent's other limitations. Integration of strained Si-Ge with NiSi-based silicide processes raises specific technical challenges—including thermal-budget constraints (excessive thermal exposure degrades the strained-Si/SiGe heterostructure), interdiffusion of germanium into the silicide-forming metal layer during anneal (which can alter silicide phase formation and disrupt the platinum gradient), and differential stress effects arising from the lattice mismatch between Si-Ge and overlying films. Tower's generic "standard component" boilerplate does not address these integration challenges, does not explain how the claimed structure would be achieved on a Si-Ge substrate with a reasonable expectation of success, and does not satisfy Tower's burden.

Tower's "Stress Liner" grouping addresses the stress-liner and oxide/oxynitride-cap limitations of claims 7-9, 11-12, and 18-19, which require, *inter alia*, a stress liner on the nickel silicide layer, a first stress liner exhibiting high compressive stress on the P-channel transistor and a second stress liner exhibiting high tensile stress on the N-channel transistor, stress liners comprising stressed silicon nitride, and an oxide or oxynitride layer on the stress liners. Tower states: "Using stress liners and other layers on the top of the transistor, like oxide or oxynitride layers, was well known and well established in the prior art" and "[s]tress liner and additional

liners are standard components of semiconductor devices that a POSITA would have been motivated to incorporate into any device.” Tower Preliminary Invalidity Contentions § V.A.3.d.

Tower has failed to carry its burden as to this grouping. Combining a stress-liner reference with a separate platinum-gradient NiSi reference requires a POSITA to have a specific reason to integrate the particular etch/overetch-resistant nickel-platinum silicide structure of the ’235 patent with the particular stress-liner material system and patterning process (including the reactive-ion-etch overetch step that the ’235 patent identifies as the source of NiSi damage). Tower’s generic observation that stress liners are “standard components” does not explain why a POSITA would have recognized the specific problem of NiSi etch damage during stress-liner patterning and would have been motivated to address that problem by formulating a platinum-gradient NiSi layer with the claimed concentration range and gradient direction. Tower has not established a reasonable expectation of success in integrating these specific teachings or a particularized motivation for doing so.

Tower’s “Exemplary Obviousness Combinations” section compounds these deficiencies. Rather than identifying a specific combination and explaining why a POSITA would have been motivated to combine those specific references with a reasonable expectation of success, Tower presents undifferentiated “either alone or in combination with . . . and/or . . . for example” permutation lists that generate hundreds of unspecified combinations. This formulation does not identify any specific combination Tower actually intends to rely upon for any specific asserted claim. It does not explain why a POSITA would select any particular permutation over any other. It does not explain how any particular combination would be implemented to yield an operable device. And it does not provide the particularized motivation-to-combine analysis required for an obviousness showing. Such shotgun arguments provide no notice of any actual theory and do not

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satisfy Tower's burden of establishing obviousness by clear and convincing evidence.

Tower has not provided a substantive patent-specific motivation-to-combine analysis for the '235 patent. Tower's "Additional Motivations to Combine" section consists of three categories—"Motivations Identified During Prosecution," "Combinations of References Through Citations," and "Combinations of Related References"—none of which supplies the required particularized showing for the specific references Tower has charted.

Tower's prosecution-history section quotes extensively from the April 26, 2012 non-final rejection issued during prosecution of the related parent application. Tower Preliminary Invalidity Contentions § V.A.5. None of these references is among the five references Tower has actually charted as primary references in this case (Shimizu '954, Akiyama '559, Choo '406, Lee '487, and Ku '182). Tower has not shown that Examiner statements addressing a different set of references supply a particularized motivation to combine the specific references Tower charts here. For example, the Examiner's reasoning as to why Okuno '047 might be combined with Dussarrat '405 or Lee 2005 or Cabral '611 does not explain why a POSITA would combine Shimizu '954 with Akiyama '559, or Choo '406 with Corni 1993, or Lee '487 with Sudo, or any other specific permutation Tower has listed in its "Exemplary Obviousness Combinations." Different references address different problems, disclose different structures, and implicate different integration challenges.

Tower's "Combinations of References Through Citations" paragraph asserts generically that "[w]here one reference cites or discusses another reference, a person of ordinary skill in the art would have been motivated to consider their teachings in combination, as they would be understood to provide related teachings in a similar field." Tower Preliminary Invalidity Contentions § V.A.5.b. Tower has not identified any instance in which any of its '235 patent

references actually cites or discusses any other charted reference. Tower identifies no common ownership between these references. Tower identifies no common inventor. The paragraph therefore provides no motivation to combine any specific references charted for the '235 patent.

Tower's "Combinations of Related References" paragraph—recycled verbatim across every patent-in-suit in this case—asserts that “[i]n some instances, multiple prior art publications and/or physical references discuss or address the same or substantially similar underlying system, software, or other project, such as commercial software products and successive versions thereof” and that “where references are owned by the same company, it would have been obvious to combine any such references.” Tower Preliminary Invalidity Contentions § V.A.5.c. This language—referring to “commercial software products and successive versions thereof”—appears to have been drafted for a software patent context and has no application to the '235 patent.

Tower's concluding *KSR* paragraph for the '235 patent states, in full: “Under *KSR*, one of ordinary skill in the art would have been motivated to combine the charted references in the specific combinations identified herein due to the fact that the references have closely related teachings, and they are addressed to the same or similar technical problem. One of ordinary skill in the art would have been motivated to combine the charted references given their related teachings of ‘sandboxing’ or restricting computer applications and resources based on user and limiting user (e.g., child user, employee, customer) access to computer applications and resources, including to limit overuse (e.g., for child use of gaming applications), prevent exposure to certain content (e.g., explicit or dangerous content), and protect user and computer security (e.g., by limiting access to potentially nefarious software).” Tower Preliminary Invalidity Contentions § V.A.5.c.

[REDACTED]

This paragraph, which purports to supply the operative *KSR* motivation-to-combine analysis for the '235 patent, discusses “sandboxing,” restricting computer applications, limiting child-user access to gaming applications, and protecting computer security. It has no relationship whatsoever to the '235 patent and appears to have been copied from an unrelated software proceeding without modification.

Tower's only remaining rationale is that claimed features were “commonly found” or “standard” in semiconductor devices. But identifying references that individually disclose claim elements is legally insufficient to establish obviousness. The Supreme Court in *KSR* rejected the rigid TSM test, but did not eliminate the requirement that a challenger show some articulated reasoning with a rational underpinning as to why a person of ordinary skill would have combined known elements in the claimed manner. Tower's “commonly found” assertions fail to provide such reasoning. They amount to the conclusory assertion that because individual elements exist in the prior art, their combination must have been obvious—precisely the hindsight-driven analysis the Federal Circuit has repeatedly condemned.

Tower lastly asserts that it offered “identical” or “materially identical” processes to those accused of infringement before the critical date of the '235 patent. Tower has not identified any such specific processes as required and therefore GlobalFoundries understands that Tower is not relying on any such processes and has waived its reliance on any such unidentified, unnamed, and unspecified processes.

Tower's failure to substantively address motivation to combine is fatal to its obviousness case as to the '235 patent. Tower has not met its burden to establish that any combination of references renders any claim of the '235 patent obvious. GlobalFoundries reserves the right to move to strike any belatedly disclosed motivation-to-combine theory and to preclude any expert

[REDACTED]

testimony that exceeds the scope of Tower's preliminary contentions.

II. U.S. Patent No. 8,507,983 ("983 Patent")

Tower has failed to meet its burden to establish by clear and convincing evidence that any prior art reference anticipates any asserted claim of the '983 patent. To anticipate, a single prior art reference must disclose each and every limitation of a claim, arranged as in the claim. Tower's thirteen claim charts (Exhibits B1 through B13) each fail to establish that the charted reference discloses all limitations of the claims to which it is mapped.

Tower has failed to meet its burden to establish by clear and convincing evidence that Shrivastava '773 discloses: a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric; a doped channel well disposed in the substrate adjacent to a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from a first heavily doped region adjacent to the first edge of the gate; a doped drift well adjacent to a second edge of the gate; a spacer on the first edge of the gate, wherein the spacer defines the displacement of the channel from the first heavily doped region; a drift isolation region in the doped drift well, and a doped deep well in the substrate of the active device region encompassing the doped drift well; the drift isolation region disposed under the gate; the first and a second heavily doped regions provided adjacent to the gate, wherein the first heavily doped region is disposed in the doped channel well on a source side of the device, and the second heavily doped region is disposed between the drift isolation region and a device isolation region on a drain side of the device; the first edge of the gate corresponding to a channel edge of the gate of the device and a second edge of the gate corresponding to a drain edge of the gate of the device; or the

[REDACTED]

the work product doctrine, and any other applicable protections. GlobalFoundries further objects to this Interrogatory as vague and ambiguous because the phrases “policy or policies,” “monitor,” and “patent-related filings” are undefined and susceptible to multiple interpretations.

In light of the foregoing objections, GlobalFoundries will not provide a response to Interrogatory No. 72.

INTERROGATORY NO. 73:

Describe all facts and circumstances related to your pre-suit investigation and preparation of the Complaint in This Investigation.

RESPONSE TO INTERROGATORY NO. 73:

GlobalFoundries incorporates all of its General Objections and reservations of rights as if set forth herein. GlobalFoundries objects to this Interrogatory to the extent it seeks information protected by the attorney-client privilege or work-product doctrine. GlobalFoundries further objects to this Interrogatory as vague and ambiguous because the phrases “facts and circumstances,” “pre-suit investigation,” and “preparation” are undefined and susceptible to multiple interpretations. GlobalFoundries further objects to this Interrogatory to the extent it seeks information that is irrelevant to the claims or defenses in this Investigation. GlobalFoundries further objects to this Interrogatory as overbroad and unduly burdensome because it seeks “all facts and circumstances” without reasonable limitation and is therefore disproportionate to the needs of this Investigation.

In light of the foregoing objections, GlobalFoundries will not provide a response to Interrogatory No. 73.

Dated: July 30, 2026

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CERTIFICATE OF SERVICE

It is hereby certified that copies of the foregoing **COMPLAINANT GLOBALFOUNDRIES U.S. INC.'S SECOND SUPPLEMENTAL OBJECTIONS AND RESPONSES TO RESPONDENTS' FIRST SET OF INTERROGATORIES (NOS. 1-73)** were served on July 30, 2026 as follows:

Stephen R. Smith COOLEY LLP 1299 Pennsylvania Ave., N.W. Suite 700 Washington, D.C. 20004 <i>Counsel for Respondents Tower Semiconductor Ltd., Tower Partners Semiconductor, Ltd., Tower Semiconductor Italy S.r.l., Tower US Holdings, Inc., Tower Semiconductor San Antonio, Inc., Tower Semiconductor NPB Holdings, Inc., Tower Semiconductor Newport Beach, Inc., and Newport Fab LLC</i>	By Email: TowerITC@cooley.com
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/s/ Erika J. Weinstein

Erika J. Weinstein

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